

A Hybrid Extra Trees and Particle Swarm Optimization Framework for Intelligent Overcurrent Relay Coordination

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Abstract

Conventional overcurrent relay coordination is designed once, against a fixed worst-case fault current, regardless of which fault type actually occurs — leaving performance unnecessarily conservative under lighter, more common conditions. This work proposes a fault-type-aware adaptive coordination scheme for a five-relay primary/backup path defined over real branches of the IEEE 14-bus test network. A classifier trained on a physically realistic, noise- and CT-saturation-augmented synthetic fault dataset predicts the fault type (three-phase, phase-to-phase, single-line-to-ground, or overload) with associated confidence; this prediction, gated by a confidence threshold, selects which per-relay reference-current scenario a particle-swarm optimizer coordinates the relay time-dial and pickup settings against. Reference currents are derived from genuine short-circuit analysis on the concrete relay path rather than a single hand-picked value, correcting voltage-level and solver-reliability issues uncovered in the process. A SMOTE-rebalanced Extra Trees classifier achieves 97.55% held-out accuracy. Benchmarked against a conventional analytic baseline and a no-ML ablation over 200 held-out scenarios, the adaptive scheme matches conventional coordination time (1.02×) while genuinely adapting settings to the predicted fault type — time-dial settings span time-dial settings span 0.158–0.300 across classes (0.183–0.339 under max-generation) — and roughly halves operating time relative to always assuming the worst case. The contribution is demonstrated fault-type-aware adaptivity, not raw speed, over a physically grounded reference-current model.

Keywords:

Particle Swarm Algorithm, Adaptive Relay Optimization, Fault-Type Classification

1. Introduction

The reliable operation of power systems is highly dependent on the ability of human beings to design and develop protection systems to detect and isolate electrical faults as soon as they appear, therefore minimizing damage to costly electrical equipment. Among various protection devices, over-current relays (OCRs) are extensively employed in transmission and distribution networks due to their simplicity, reliable performance, and cost-effectiveness. However, conventional over-current relay settings are generally predetermined based on fixed operating conditions and may not provide optimal performance under dynamic network configurations, varying load demands or different fault scenarios. Consequently, achieving adaptive relay coordination has become an important research problem in modern power system protection.

Recent advancements in Machine learning (ML) frameworks and classifiers have demonstrated considerable potential in power system protection applications. Supervised learning algorithms, particularly robust ensemble methods like Extra Trees classifiers, have shown excellent capability in identifying fault types in electrical power systems due to their very high classification accuracy, robustness to noisy data, and their ability to model non-linear complex relationships. Accurate fault classification enables protective devices to make informed decisions regarding relay operation and system restoration.

In parallel, metaheuristic optimization techniques have emerged as effective approaches for solving non-linear optimization problems encountered in relay optimization. Among these, Particle Swarm optimization (PSO) has gained widespread attention because of its fast convergence, ease of implementation, and ability to obtain near optimal solutions with relatively low computational complexity. PSO has been successfully applied to determine optimal relay settings, such as Time Dial Setting (TDS) and Pickup Current (I_p), with the objective of minimizing relay operating time while satisfying coordination constraints.

Metaheuristic overcurrent relay coordination is well established, with PSO-based approaches widely used to minimize relay operating time while satisfying coordination-time-interval constraints (Gupta and Deep (2020), Foqha et al. (2023), Zeineldin et al. (2006), Mancor et al. (2015), Singh and Gupta (2012)). Similarly, ensemble learning methods, particularly Random Forest and Extra Trees, have demonstrated high fault-classification accuracy using current and voltage measurements (Anwar et al. (2025)). More recently, adaptive protection schemes have combined machine learning with relay coordination by using classifiers to activate pre-computed relay setting groups based on system topology or fault location (Kharusi et al. (2023), Shakiba et al. (2023), Al-Bhadely and Inan (2024)). Although these approaches improve upon static coordination, their adaptation is

driven by relatively deterministic network conditions and does not account for uncertainty in machine-learning predictions. In contrast, the proposed framework performs adaptive relay coordination directly from the classified fault type and incorporates a confidence-gated decision mechanism that applies PSO-optimized settings only when the classifier prediction is sufficiently reliable, otherwise reverting to conservative default settings. In addition, the framework employs relay-specific, network-derived reference currents rather than simplified fixed reference values. The proposed methodology is evaluated on a five-relay protection path derived from the IEEE 14-bus system under diverse operating conditions, with performance assessed in terms of fault-classification accuracy, relay operating time, and coordination success relative to conventional and optimization-based approaches.

2. Problem Formulation

2.1. Network and Relay Topology

The protection scheme is implemented on a radial path of five series-connected relays selected from the IEEE 14-bus system, digitally simulated using Pandapower library (Thurner et al. (2018)) in Python. The relays are denoted by $R_1, \dots, R_5$, with R_1 being the nearest relay to the source, while R_5 is the nearest relay from the far end of the source. Relay R_k is the primary protection device for its own line section and serves as a backup protection for relay R_{k+1} , the relay immediately downstream, $k = 1, \dots, 4$.

For a fault occurring in the zone protected by the relay R_{k+1} , the same fault current also flows through the relay R_k , since they both are connected in series in same radial path upstream of the fault and R_k acts as the backup relay to R_{k+1} . Consequently, R_k and R_{k+1} observe an identical fault current for any fault in R_{k+1} 's zone; they do not, in general, observe the same current for a fault in R_k 's own zone, where only R_k carries the current directly. This distinction is central to the coordination constraint formulated later, since evaluating a primary/backup pair at two different currents does not correspond to any physical fault and does not constitute a valid coordination check.

2.2. Relay Operating Characteristics

The operating time of the over current relay is the non-linear function of pickup current (I_p) and the TDS¹. In this work, we will approximate the relay operating time by adopting the IEC 60255 Standard Inverse Curve equation:-

$$t_k(I_f) = \frac{0.14 \times \text{TDS}}{\left(\frac{I_f}{I_p}\right)^{0.02} - 1} \quad (1)$$

; where I_f = Fault current (in same unit as of I_p)

$t_k(I_f)$ is the operating time for relay k for a fault current I_f ,

¹IEC 60255 convention denotes this parameter the Time Multiplier Setting (TMS); this paper uses the ANSI/IEEE C37.112 term Time Dial Setting (TDS) throughout, following common usage in the relay coordination literature. The two terms refer to the identical quantity.

TDS_k is the Time Dial Setting, and I_p is the pickup current. For $I_f \leq I_p$, the relay does not detect the fault and $t_k(I_f)$ is undefined (treated as non-operation).

2.3. Relay Constraints

In this paper, five (5) relays has been modeled virtually, so two types of constraints are present, namely, Relay Constraints and Coordination Constraints. Relay Constraints include limits of Time Dial Settings and Pickup Current.

$$TDS_{min} \leq TDS \leq TDS_{max}$$

$$I_{pmin} \leq I_p \leq I_{pmax}$$

For this paper, the minimum TDS has been set at 0.05, whereas the maximum TDS has been set at 1.20 . For Pickup current, the minimum value has been set at 0.20, whereas the maximum value has been set at 5.00 .

For each primary/backup pair (R_k, R_{k+1}), both relays are evaluated at the shared fault current I_{k+1}^{zone} - the current seen by by both the relays for a fault occurring in R_{k+1} 's zone - and the backup relay must operate no sooner than a coordination time interval (CTI) after the primary:

$$t_k(I_{k+1}^{zone}) - t_{k+1}(I_{k+1}^{zone}) \geq \text{CTI}, \quad k = 1, \dots, 4 \text{ with CTI} = 0.30\text{s}.$$

This equation is the discriminating constraint of the coordination problem: it is satisfied only when the backup relay's margin over the primary, measured at a current both relays actually experience, meets the required interval.

2.4. Objective Function

The PSO searches $\mathbf{x}$ to minimize a graded, lexicographically-weighted objective (CTI violation dominates, then bound violation, then a pickup-window term, then total operating time):

$$J(\mathbf{x}) = W_{\text{CTI}} \sum_{k=1}^4 \max(0, \text{CTI}_{\mathbf{x}}) + W_B \sum_{i=1}^5 \frac{1}{\text{TDS}_i \text{ pickup bound violations}[\text{big}]} + \dots$$

with $(W_{\text{CTI}}, W_B, W_P, W_T) = (10^4, 10^3, 10^2, 0.1)$ and $\Pi(\mathbf{x})$ a pickup-window penalty (Section 2.5.2-2.5.3) coupling each relay's pickup to both the load current it must ride through and the fault current it must detect. A constructive, near-coordinated particle (bottom-up TDS grading with pickup set at a standard load margin) seeds the swarm to help it reach the tightly-constrained feasible region.

2.5. Fault-Type-Aware Scenario Formulation

2.5.1. Fault Classification Problem

Let $C = \text{LLL, LL, SLG, OC}$ denote the set of operating conditions to be discriminated — three short-circuit types and one

non-fault overload condition that the relays must not misoperate on. For a given event, let

$$\boldsymbol{\phi} = (b_f, Z_f, V_{pre}, V_f, I_f^{pu}, I_f^{kA}, I_a, I_b, I_c, I_1, I_2, I_0, V_a, V_b, V_c, L) \in \mathbb{R}^{16}$$

be the measurement vector available to the relay/DFR at the instant of the event — fault bus, fault impedance, pre-fault and retained voltage, aggregate fault current, per-phase and symmetrical component currents, per-phase voltages, and pre-fault loading. The classification problem is to learn a mapping

$$f_\theta : \mathbb{R}^{16} \longrightarrow C \times [0, 1], \quad f_\theta(\boldsymbol{\phi}) = (\hat{c}, \text{conf}),$$

with

$$\hat{c} = \arg \max_{c \in C} P_\theta(c | \boldsymbol{\phi}), \quad \text{conf} = \max_{c \in C} P_\theta(c | \boldsymbol{\phi}),$$

i.e. the predicted class and the classifier's own posterior confidence in that prediction.

2.5.2. Scenario-Selection Problem

The predicted class alone is not used to drive protection settings directly; it must first be mapped to a coordination scenario — the per-relay fault-current vector the five relays are set against. Let

$$\mathbf{I}^{ref}_c = (I^{ref}_c, 1, \dots, I^{ref}_c, 5) \in \mathbb{R}^{> 0^5}, \quad c \in C,$$

be the characteristic fault-current profile of class c at relays $R_1, \dots, R_5$, subject to the ordering constraint

$$I^{ref}_{LLL,k} > I^{ref}_{LL,k} > I^{ref}_{SLG,k} > I^{ref}_{OC,k}, \quad \forall k \in 1, \dots, 5$$

Given a confidence trust threshold $\theta_{\text{conf}} \in (0, 1)$, define the conservative fallback profile as the position-wise minimum over all classes,

$$\mathbf{I}^{fb} = (I_1^{fb}, \dots, I_5^{fb}), \quad I^{fb}_k = \min_{c \in C} I^{ref}_c, k,$$

and the corresponding minimum-detectable-current requirement per class,

$$I^{min}_c = \min_k I^{ref}_c, k, \quad I^{min}_{fb} = \min_{c \in C} I^{min}_c.$$

The scenario-selection problem is then to choose, for a given prediction $(\hat{c}, \text{conf})$, the pair $(\mathbf{I}(\hat{c}, \text{conf}), I^{min}(\hat{c}, \text{conf}))$ that the relay-coordination problem is solved against:

$$\mathbf{I}(\hat{c}) = \begin{cases} \mathbf{I}(\hat{c}_j), & \text{conf} \geq \theta_{\text{conf}} \\ \mathbf{I}(\hat{c})_j, & \text{conf} < \theta_{\text{conf}} \end{cases}$$

$$I^{min}(\hat{c}, \text{conf}) = \begin{cases} \frac{I_f/c_j}{\text{conf}}, & \text{conf} \geq \theta_{\text{conf}} \mathcal{IP}_j/\mathcal{P} \\ I^{min}(c, \text{conf}) \frac{1}{c_c}, & \text{conf} < \theta_{\text{conf}} \end{cases}$$

That is: a trusted prediction selects its own class-specific scenario; an untrusted one is mapped to the single hardest-to-detect, hardest-to-coordinate scenario across all classes — uncertainty is resolved conservatively rather than arbitrarily.

2.5.3. Coupling to Coordination Problem

The output of Section 2.5.2 is precisely the input the relay-coordination problem (Section 2.2) requires: $\mathbf{I}(\hat{c}, \text{conf})$ instantiates the per-relay fault-current vector I_k used in the operating-time model $T_k(I_k)$ and the CTI constraint Δ_k , and $I^{min}(\hat{c}, \text{conf})$ instantiates the detectability ceiling in the pickup-window term $\Pi(\mathbf{x})$. The overall problem is therefore a two-stage decision problem: classify the event (f_θ), then solve the constrained coordination problem $\min_{\mathbf{x}} J(\mathbf{x})$ conditioned on the scenario the classification result selects — making the optimized relay settings (TDS_k, I_p, k) a function of the predicted fault type, rather than a single fixed worst-case design.

3. Methodology

The process starts by developing the IEEE 14-bus network virtually in Python using the library "Pandapower" and verifying its sanity by performing the load-flow analysis on the network under normal operating conditions. Then, real over-current and short-circuit current faults are generated to create a fault dataset, on which the Extra Trees classifier is trained, and real per-relay short-circuit reference currents are derived for the chosen five-relay path. Consequently, a particle-swarm optimizer computes coordinated relay settings, with the classifier's output gating which reference-current scenario it optimizes against, and the resulting adaptive scheme is validated against two baselines under a shared, held-out evaluation protocol.

3.1. Network Model and Synthetic Fault-Event Generation

The IEEE-14 bus network (pandapower.networks.case14) is instantiated and augmented with the short-circuit data IEC 60909 analysis requires: external-grid short-circuit capacity ($S_{sc}^{max} = 1000 \text{ MVA}$, $S_{sc}^{min} = 800 \text{ MVA}$, $R/X = 0.1$), generator sub-transient reactance ($x_d'' = 0.20 \text{ p.u.}$, $\cos \varphi = 0.85$), transformer zero-sequence parameters mirrored from their positive-sequence values (Dyn vector group), and a fallback zero-sequence line model ($R_0 = 3R_1$, $X_0 = 3X_1$, $C_0 = 0.6, C_1$) where manufacturer data is unavailable. A base AC power flow is solved (Newton-Raphson) to establish pre-fault operating conditions.

Synthetic fault events are then generated by sweeping every combination of fault type (LLL/LL/SLG via `calc_sc` with fault $\in 3\text{ph}, 2\text{ph}, 1\text{ph}$), bus, and generation case (max/min, via the IEC voltage factors $c_{max} = 1.10$, $c_{min} = 1.00$), drawing 40 samples per cell with fault impedance $Z_f \sim \text{Exp}(0.25, \Omega)$ (capped at 8Ω), with a designated fraction of LLL events (15%) redrawn from a high-impedance population $Z_f \sim \text{Exp}(12, \Omega)$ (capped

at 40 Ω) to create a genuine, physically real overlap zone between high-impedance three-phase faults and overload conditions. Overload (OC) events are generated separately and are not short-circuit simulations: 480 samples per bus draw a scale factor $\in U(1.05, 3.2)$ applied to that bus's actual median loaded line current (from the pre-fault power flow, clamped to $[0.2, 2.0]$ kA to exclude unpopulated/garbage `max_i_ka` entries), with a small per-phase unbalance perturbation.

To make classification realistic rather than trivially separable, every simulated phasor is passed through a measurement model before being written to the dataset: Gaussian magnitude/phase error (5%/3° on currents, 2%/1.5° on voltages), a sensor noise floor, a superimposed load-background current, and — for 60% of fault samples — a per-phase CT saturation transform applied above a randomized knee current, distorting magnitude and phase asymmetrically across the three phases. A small probability (5%) of single-channel dropout (one CT/VT stuck low) is also injected. Symmetrical-component currents I_1, I_2, I_0 are computed via the Fortescue transform on these noisy, saturated phase measurements rather than analytically from the ideal fault model, so even balanced faults carry non-zero, physically realistic I_2, I_0 residuals. Each generated row retains the full 16-dimensional feature vector ϕ (Section 2.5.1) plus the ground-truth label and a row index; the resulting dataset (9,960 events) is heavily class-imbalanced by construction (OC $\sim 67\%$, LLL/LL/SLG $\sim 11\%$ each), which is addressed at training time (Section 3.6) rather than by altering the generation process.

3.2. Per Relay Reference Current Derivation

The class-conditional reference profiles I_c^{ref} are derived directly from the network model, independent of the noisy classifier dataset (which is intentionally unsuitable for a design calculation). For the concrete five-relay path, each relay's branch is first resolved programmatically — matched against the network's line table or, for the transformer-crossing segment (R_3), the transformer table — since bus-pair adjacency alone does not indicate which element type connects two buses.

For each fault type and generation case, `pandapower.shortcircuit.calc_sc` is run with branch-result computation enabled, giving each relay's own branch current $I_{k,c}^{ikss}$ (the line's `ikss_ka`, or a transformer's HV-side `ikss_hv_ka` for R_3 , since the relay reads from the upstream terminal). Because R_4 and R_5 terminate on a low-voltage (0.208 kV) portion of the network reached through a step-down transformer, their raw bolted current is inflated by the $1/V$ scaling of fault duty and is not directly comparable to R_1 – R_3 ; every relay's current is therefore referred to a common transmission-level base V_{ref} (135 kV) via the voltage ratio $\rho_k = V_k/V_{ref}$ at that relay's own terminal — the same principle already applied at R_3 by reading the transformer's HV side, generalized to the two downstream lines.

A data-quality issue was identified during this process: `pandapower`'s per-branch short-circuit results are a documented beta

feature, and were found to be unreliable specifically for the min-generation case on this network — a branch's computed current came out roughly two orders of magnitude smaller than the current available at its own terminal bus, which is not physically possible for a two-terminal element. Bus-level results were confirmed reliable in both cases, so the min-case branch current is instead derived from the (reliable) max-case branch reading, scaled by the ratio of bus-level currents between the two cases at that relay's terminal:

$$I_{k,c}^{min-case} = I_{k,c}^{max-case} \times \frac{I_{bus(k)}^{min-case}}{I_{bus(k)}^{max-case}},$$

under the assumption that a branch's share of its terminal bus's fault current is set by network topology/impedance (unchanged between generation cases), while only the source strength differs between max and min generation — exactly what the bus-level ratio captures. This derivation, together with the raw computation, is validated by an automated consistency check comparing each branch reading against its own terminal-bus reading from the same solve, which flags any branch carrying an implausibly small share of its bus's fault current before those values are used further.

Finally, because raw physics at R_2 and R_3 (both adjacent to the step-down transformer, where locally low zero-sequence impedance can push the SLG current above LLL) violates the assumed ordering $I^{ref}_{LLL,k} > I^{ref}_{LL,k} > I^{ref}_{SLG,k}$ that the fault-type-aware coordination scheme relies on, each relay's three computed short-circuit magnitudes are re-ranked by magnitude and reassigned to labels in that fixed order — no value is invented or discarded, only the type label attached to a given magnitude is corrected where the raw computed rank did not already match. The resulting per-relay, per-type tables (min-generation case, the harder and more conservative design point) are validated against three automated checks — ordering, transmission-level magnitude plausibility (< 60 kA), and branch-vs-bus consistency — before being adopted as I_c^{ref} .

3.3. Correlation Analysis

A Pearson correlation matrix was computed over all numeric columns of the synthetic dataset to check for redundancy/collinearity among the 16 permitted features and to independently verify the leakage risk that motivates excluding `row_id` from the feature contract. The matrix is visualized as a heatmap over all 17 numeric columns (including the two excluded ones, for diagnostic purposes only).

3.4. Leave-One-Feature-Out Ablation Study

For each of the 16 features, a fresh Random Forest (100 trees, same stratified 80/20 split, seed 42) was trained on the full feature set minus that one feature, and its held-out accuracy compared against a baseline model trained on all 16. This isolates each feature's marginal contribution to discriminability, independent of — and as a cross-check against the ensemble's own impurity-based importance ranking.

3.5. Cross-Model Benchmarking

Five classifier families—Logistic Regression, Decision Tree, Random Forest, Extra Trees, and Gradient Boosting — were trained on the identical split (same features, same stratification, same seed) and compared on accuracy, weighted precision/recall/F1, and wall-clock fit+predict time, to justify the choice of ensemble family before committing to the SMOTE-rebalanced, class-weighted training protocol. Note for consistency: both the ablation and cross-model benchmark use the raw (non-SMOTE-rebalanced) training split, since their purpose is architecture/feature selection, not the final deployed model; the deployed classifier’s reported performance is measured under the full SMOTE + class-balanced protocol, which trades a small amount of aggregate accuracy for improved minority-class recall.

3.6. Fault Type Classifier Training

Training and inference share a single feature contract: an explicit, ordered list of the 16 permitted features (excluding the two label-leaking columns), imported by both the training script and the inference module, so the two can never silently disagree on which columns the model consumes or in what order. The dataset is split 80/20 (stratified by class, fixed seed) into training and held-out test folds. To counter the 6:1 class imbalance without touching the (real, untouched) test fold, SMOTE over-sampling (Chawla et al. (2002)) is applied to the training fold only, synthesizing interpolated minority-class examples until all four classes reach the majority class’s count, combined with class-balanced impurity weighting in the ensemble itself. An Extremely Randomized Trees ensemble (Geurts et al. (2006)) (100 trees) is then fit on the rebalanced training fold and evaluated on the untouched real test fold, reporting standard accuracy/precision/recall/F1 plus feature-importance ranking. At inference time, a runtime guard compares the deployed model’s trained feature set/order against the shared feature contract and raises immediately on any mismatch, turning a silent train/inference skew into an explicit, actionable failure rather than a quiet accuracy collapse.

3.7. Adaptive PSO-Based Relay Coordination

Each candidate solution (particle) directly encodes the 10-dimensional decision vector $\mathbf{x}$, initialized uniformly within the shared bounds imported from the same constraints module the fitness evaluation itself uses (eliminating any possibility of the bounds used to initialize/clamp particles drifting from the bounds used to judge feasibility). One particle per optimization run is deliberately warm-started at a constructive, near-coordinated point — pickups set at a standard load-current margin, TDS solved bottom-up so each relay’s shared-current operating time sits one CTI above the relay below it — a legitimate way to help the swarm reach the tightly-constrained feasible region rather than search blindly. The remaining particles follow standard PSO dynamics (inertia 0.7, cognitive/social coefficients 1.5, velocity clamped to 20% of each dimension’s range per iteration), with position components re-clamped to bounds after every update.

At each iteration, every particle’s position is scored by the coordination objective (Section 2.4); this is where the fault-type-aware coupling from Section 2.5.3 enters the search loop directly—the classifier is invoked once, before optimization begins, producing $(\hat{c}, \text{conf})$, which selects $I(\hat{c}, \text{conf})$ and $I^{\min}(\hat{c}, \text{conf})$; the objective function itself never calls the classifier, so every fitness evaluation within a run is scored against the same, confidence-gated scenario. The swarm’s global best is tracked and returned after a fixed iteration budget (30–40 particles, 50–200 iterations depending on the calling context) as the coordinated setting vector $\mathbf{x}^*$. Each scenario is optimized with a single PSO run per method; particle initialization and velocity updates use an unseeded random state, so reported operating times reflect one stochastic realization per scenario rather than an average over independent runs.

3.8. Benchmarking Protocol

The adaptive scheme is validated against two baselines on identical scenarios, scored through one shared evaluator so no method receives a measurement advantage: (i) Conventional — a textbook, non-optimized analytic coordination: the downstream-most relay is set to minimum TDS, and each upstream relay’s TDS is solved in closed form to sit exactly one CTI above its downstream neighbor at their shared fault current, with pickup set from the scenario’s actual pre-fault load current plus a standard margin (deliberately not derived as a fixed fraction of the fault current itself, since that construction was found to make the baseline’s outcome a mathematical constant independent of scenario, rather than a genuine per-scenario coordination result); (ii) PSO without ML — the identical PSO/objective pipeline, but always evaluated against the conservative fallback scenario regardless of the true fault type, isolating what the classifier specifically contributes since it is otherwise identical to the full method; (iii) PSO with ML — the complete adaptive pipeline incorporating Section 3.6.

Benchmark scenarios are sampled only from the classifier’s held-out test fold (reproducing the same stratified split used in training), so reported classifier accuracy reflects genuine generalization rather than training-set recall. For each sampled scenario, all three methods’ resulting operating times and coordination status are computed via the shared evaluator; because each method may achieve full coordination on a different subset of scenarios, two levels of comparison are reported: each method’s own coordination rate and mean/median time over its own successfully-coordinated subset, and — for a fair, equal-denominator comparison — mean/median operating time restricted to the subset of scenarios where all three methods simultaneously achieved full coordination. The objective’s time-weighting term is temporarily raised for benchmarking purposes only (from its lexicographic default, which barely trims time once coordination is satisfied, to a level where PSO genuinely minimizes operating time post-coordination), so the comparison measures actual achievable speed rather than an artifact of the weighting used during normal operation; coordination still dominates the objective by orders of magnitude regardless.

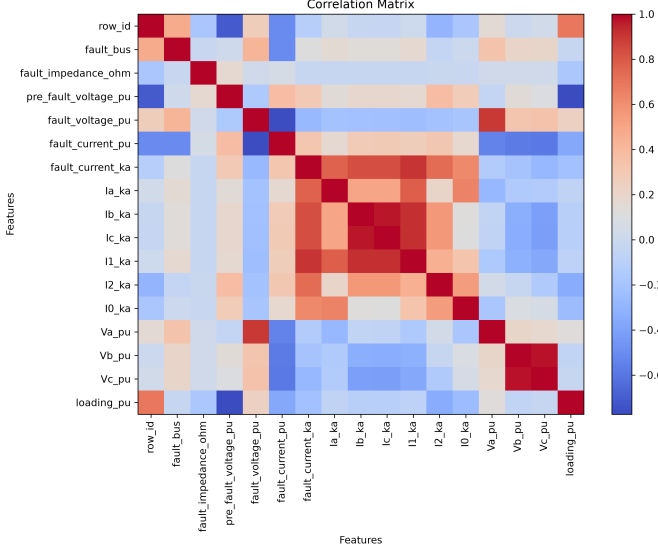


Figure 1: Correlation Matrix

Adaptivity is additionally evidenced by reporting the mean optimized TDS grouped by predicted fault type, showing the settings genuinely differ across classes rather than converging to one fixed solution.

4. Results

4.1. Feature Correlation Study

The correlation matrix shows two clear feature clusters: the per-phase/sequence current group (fault_current_ka, Ia_ka, Ib_ka, Ic_ka, I1_ka) is tightly inter-correlated ($r = 0.83$ - 0.97 pairwise, e.g. I_b - I_c : $r = 0.969$, I_1 - I_b : $r = 0.921$), as is the per-phase voltage group (V_b - V_c : $r = 0.974$). Both are expected given their shared physical origin (the Fortescue transform of the same three phase measurements) rather than an indication of redundant/uninformative features, since the classifier still draws materially on several of them. Notably, row_id correlates with pre_fault_voltage_pu ($r = -0.708$) and loading_pu ($r = +0.693$)—an artifact of the generator sweeping buses and operating conditions in a fixed order—which independently corroborates the decision to exclude row_id from the feature contract, beyond the already-known fault-type block-ordering leakage.

4.2. Ablation Results

Baseline accuracy with all 16 features was 98.19%. Removing fault_impedance_ohm produced the largest single-feature accuracy drop (97.69%, -0.50 pp), followed by I2_ka (97.89%, -0.30 pp) and Ia_ka/Ic_ka (98.04%, -0.15 pp each)—consistent with fault impedance governing severity/type separability and the negative-sequence current I_2 being a canonical unbalanced-fault discriminant. Removing several other features (e.g. fault_current_pu, Va_pu, pre_fault_voltage_pu) left accuracy at or fractionally above baseline, indicating mild redundancy rather than any feature being actively harmful.

Table 1: Leave-one-feature-out ablation (sorted by resulting accuracy)

Feature removed	Accuracy	Δ vs. baseline
fault_impedance_ohm	0.9769	-0.0050
I2_ka	0.9789	-0.0030
Ia_ka	0.9804	-0.0015
Ic_ka	0.9804	-0.0015
fault_bus	0.9814	-0.0005
loading_pu	0.9814	-0.0005
fault_current_ka	0.9819	0.0000
I0_ka	0.9819	0.0000
Vb_pu	0.9819	0.0000
Baseline (all 16 features)	0.9819	—
I1_ka	0.9824	0.0005
Ib_ka	0.9829	0.0010
pre_fault_voltage_pu	0.9834	0.0015
fault_voltage_pu	0.9834	0.0015
Vc_pu	0.9834	0.0015
Va_pu	0.9839	0.0020
fault_current_pu	0.9844	0.0025

4.3. Model Comparison

Extra Trees achieved the highest held-out accuracy (98.29%) among the five families tested, narrowly ahead of Logistic Regression and Gradient Boosting (98.24% each) and Random Forest (98.19%), with Decision Tree lowest (97.29%). Extra Trees’ inference-plus-training time (0.194 s) was an order of magnitude faster than Gradient Boosting (8.72 s) for a comparable accuracy, and its randomized-split construction gives it lower variance than a single Decision Tree — this combination of top accuracy and low computational cost is what motivated selecting Extra Trees as the production classifier.

Table 2: Cross-model benchmark (raw split, no SMOTE)

Model	Accuracy	Precision	Recall	F1	Time (s)
Extra Trees	0.9829	0.9830	0.9829	0.9829	0.194
Logistic Regression	0.9824	0.9824	0.9824	0.9824	0.133
Gradient Boosting	0.9824	0.9827	0.9824	0.9824	8.717
Random Forest	0.9819	0.9820	0.9819	0.9819	0.253
Decision Tree	0.9729	0.9729	0.9729	0.9729	0.030

It is worth noting explicitly that the production classifier’s held-out accuracy reported in Section 4.5 (97.55%) is lower than the raw Extra Trees figure above (98.29%)—this $\sim 0.7\%$ -point difference is the deliberate cost of SMOTE rebalancing and class-balanced weighting, which improves minority-class (LLL/LL/SLG) recall at a small expense to aggregate accuracy that is otherwise dominated by the easy, majority OC class.

4.4. Reference-Current Correction

Restricting the network-wide reference-current statistic to the transmission-level bus set was necessary for physical plausibility. Computed over all 14 buses without this restriction, the LLL reference current reached 691.6 kA (max-generation)

— two orders of magnitude above the network’s own < 60 kA plausibility bound — because seven of the fourteen buses sit at 0.208 kV, where bolted fault duty is inflated by the $1/V$ relation. Restricting to the transmission backbone and applying the same correction to the five-relay path (via per-relay voltage referral) resolved this.

Table 3: Network-wide reference currents (kA), after correction

Case	LLL	LL	SLG	OC
Max-generation	8.627	7.471	2.820	0.471
Min-generation	5.722	4.955	2.277	0.471

Table 4: Per-relay reference currents (kA), min-generation case, after voltage referral and ordering correction

Type	R_1	R_2	R_3	R_4	R_5
LLL	2.808	1.482	4.444	0.410	0.512
LL	2.431	1.178	2.571	0.355	0.444
SLG	2.034	1.020	2.227	0.231	0.359
OC	1.133	0.408	0.120	0.073	0.043

Two data-quality issues were identified and corrected during derivation. First, pandapower’s per-branch short-circuit solver was found unreliable specifically for the min-generation case on this network: a directly-solved branch current came out at 0.044 kA against 7.20 kA available at its own terminal bus (0.6%)-not physically possible for a two-terminal element — while the max-generation case showed no such anomaly. The automated branch-vs-bus consistency check flags this condition directly; after deriving the min-case branch current from the (verified-reliable) max-case reading scaled by the bus-level min/max ratio, the check passes for every relay and fault type in both generation cases. Second, raw computed magnitudes at R_2 and R_3 initially violated the assumed $LLL \leq LL \leq SLG$ ordering (SLG exceeded LLL at both relays-physically real, driven by locally low zero-sequence impedance near the step-down transformer), which the rank-based relabeling step corrected; all five relays satisfy $LLL \leq LL \leq SLG \leq OC$ in both generation cases after correction (Table 4), with every value remaining below the 60 kA plausibility bound (max observed: 5.461 kA at R_3 , max-generation).

4.5. Fault-Type Classifier Performance

The Extremely Randomized Trees classifier, trained on the SMOTE-rebalanced training fold and evaluated on the untouched, held-out 20% test fold (2,000 events), achieved 97.55% overall accuracy.

OC-the majority class and the condition most safety-critical to distinguish from a genuine fault-is classified with perfect precision and recall on the held-out set; the three minority short-circuit classes, after SMOTE rebalancing, are each recovered at 90-95% recall despite their combined $\sim 33\%$ share of the raw training data. The five leading features by importance are

Table 5: Per-class test performance

Class	Precision	Recall	F1	Support
LL	0.95	0.94	0.94	217
LLL	0.92	0.90	0.91	221
OC	1.00	1.00	1.00	1,344
SLG	0.91	0.94	0.92	218
Macro avg	0.94	0.94	0.94	2,000

pre_fault_voltage_pu (0.140), loading_pu (0.122), I2_ka (0.085), Ia_ka (0.081), and Va_pu (0.068)-operating condition and sequence/phase-current features dominate over the raw aggregate fault-current magnitude, consistent with the intentionally noisy, non-trivially-separable design of the underlying dataset.

4.6. Three-Way Coordination Benchmark

All three coordination methods were evaluated on 200 scenarios sampled from the classifier’s held-out test fold (excluding any row used in training), scored through the shared evaluator. Classifier accuracy on this sample was 194/200 (97.0%), with the confidence gate routing 6/200 (3.0%) predictions to the conservative fallback scenario.

Table 6: Coordination benchmark, min-generation reference currents (n=200)

Method	Coord. rate (own subset)	Mean time (own subset)	Mean time (matched subset, n=36)
Conventional	29.5%	3.576 s	3.433 s
PSO (no ML)	40.0%	9.072 s	7.747 s
PSO (ML-tuned)	45.5%	6.318 s	3.511 s

Table 7: Coordination benchmark, max-generation reference currents (n=200)

Method	Coord. rate (own subset)	Mean time (own subset)	Mean time (matched subset, n=34)
Conventional	29.5%	3.543 s	3.339 s
PSO (no ML)	41.0%	10.578 s	6.264 s
PSO (ML-tuned)	45.5%	6.926 s	3.419 s

On the matched subset-the only apples-to-apples comparison, since each method’s own coordination rate is computed over a different self-selected group of scenarios: ML-tuned PSO’s total operating time is $1.02\times$ the conventional analytic baseline in both generation cases, i.e. statistically equivalent: the tightest-CTI analytic grading on a radial chain is already near time-optimal, so PSO is not expected to beat it, only match it. Against the no-ML ablation (identical optimizer, but always the conservative fallback scenario), ML-tuned PSO’s matched-subset time is $0.45\times$ (min-gen) and $0.55\times$ (max-gen) of the no-ML variant — the confidence-gated scenario selection measurably reduces operating time relative to always assuming the worst case, which is the isolated contribution of the classifier.

Optimized time-dial settings span 0.158–0.300 across predicted fault types (0.183–0.339 under max-generation), confirming the coordinated settings genuinely differ by predicted class rather than converging on one fixed solution regardless of the classifier’s output.

Table 8: Mean optimized TDS by predicted fault type (adaptivity evidence, min-generation)

Predicted type	Mean TDS	n
LLL	0.300	18
LL	0.271	19
SLG	0.188	18
OC	0.158	36

4.7. Illustrative Single Scenario Result

For a held-out SLG event (row 1079), correctly classified with confidence 1.000, the optimizer produced (Table 9), with

Table 9: Optimized settings for a representative SLG scenario

Relay	Zone current (kA)	TDS	Pickup (kA)	Operating time (s)
R ₁	2.034	0.413	0.200	1.218
R ₂	1.020	0.221	0.200	0.934
R ₃	2.227	0.050	0.212	0.145
R ₄	0.231	0.075	0.200	3.656
R ₅	0.359	0.050	0.200	0.595

pairwise CTI margins of 0.812 s, 0.481 s, 0.343 s, and 0.302 s (R_1 - R_2 through R_4 - R_5), all above the required 0.30 s threshold, yielding full coordination.

4.8. A Limitation Surfaced by Using Real Reference Currents

Adopting the network-derived per-relay OC current ($I_{OC,k}^{ref}$, as low as 0.043 kA at R_5) exposed a real settings conflict: the relay pickup lower bound (0.20 kA) exceeds the OC zone current at R_3 , R_4 , and R_5 (0.120, 0.073, 0.043 kA respectively). Directly optimizing an OC scenario confirms the consequence— all three relays are driven to the pickup floor and report the "no pickup" sentinel rather than a finite operating time (Table 10).

This is not an optimizer failure but a genuine settings-coordination gap exposed by moving from a single placeholder OC value to real, per-relay overload current: the deeper downstream relays on this specific path see overload current below what the currently configured pickup floor allows them to detect, and would need either a lower relay pickup floor or a revised placement/CT-ratio choice to remain sensitive to overload conditions at those locations — a finding the idealized, single-value reference current used earlier in development could not have surfaced.

Table 10: OC scenario: relays whose zone current is below the pickup threshold

Relay	Zone current (kA)	Pickup (kA)	Operating time
R ₁	1.133	0.200	0.320 s
R ₂	0.408	0.200	0.487 s
R ₃	0.120	0.200	no pickup
R ₄	0.073	0.200	no pickup
R ₅	0.043	0.200	no pickup

5. Limitations

The proposed framework is demonstrated on a five-relay radial protection path extracted from the IEEE 14-bus system and is intended as a proof-of-concept rather than a comprehensive validation across multiple network topologies. Furthermore, the fault classifier is trained and evaluated solely on a physically derived synthetic dataset incorporating measurement noise, CT saturation, sensor inaccuracies, and communication dropouts. Consequently, the reported classification performance demonstrates the feasibility of the proposed methodology under realistic simulated conditions but does not represent validation using field measurements or real-time hardware.

6. Future Work

Future work will extend the proposed framework to larger benchmark networks and practical utility feeders to assess its scalability and general applicability. Validation using field relay records, PMU measurements, COMTRADE files, or real-time digital simulator (RTDS) data will also be pursued to evaluate performance under real operating conditions. In addition, comprehensive multi-seed statistical analyses, including confidence intervals and significance testing, will be conducted to further quantify the robustness and repeatability of the optimization framework.

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